

# UGLP: Unifying Global and Local Preferences for Multi-behavior Recommendation

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**Abstract**—Multi-behavior recommender systems have demonstrated their effectiveness in mitigating issues such as data sparsity by incorporating auxiliary behaviors into the target behavior. However, existing multi-behavior recommendation approaches typically take one of two directions: (1) fusing behavior-specific preference features from various behavior interaction graphs explicitly or implicitly for recommendation; or (2) utilizing behavior-unified preference features from the unified interaction graph for recommendation or to initialize features for subsequent modeling. These methods fail to exploit the integration of behavior-unified global and behavior-specific local preference features, resulting in incomplete preference modeling. To address this issue, in this work, we propose a novel method called Unifying Global and Local Preferences (UGLP) for multi-behavior recommendation. In UGLP, we design a behavior feature fusion network that consists of global and local fusion modules for comprehensive and fine-grained user preferences. The global fusion module performs graph convolution on behavior-unified global and behavior-specific local interaction graphs to obtain behavior-unified and behavior-specific features. The behavior-unified and behavior-specific features are then fused into globally fused features via a gating network. The local fusion module then performs cross-behavior fusion on these globally fused features via another gating network. We introduce a contrastive learning module to promote preference alignment and knowledge transfer from auxiliary behaviors to the target behavior. Additionally, we incorporate a GCN refinement module to adjust the fused features to ensure that both global and local user preferences are learned. Experimental results on three real-world datasets verify that our method is able to surpass various state-of-the-art models. For instance, our method outperforms the best baseline by an average of 16.43% and 14.36% in terms of HR@10 and NDCG@10, respectively.

**Index Terms**—Collaborative filtering, multi-behavior recommendation.

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## I. INTRODUCTION

RECOMMENDER systems have already been an effective mechanism to leverage user preferences, assisting individuals in discovering items tailored to their personalized interests. These systems have become integral technologies across a wide range of platforms. Collaborative filtering (CF) [1], [2], one of the most successful strategies, has evolved from shallow models [2], [3], [4] to deep neural models [5], [6], [7]. Recently, Graph Convolutional Networks (GCNs) have been introduced to capture high-order connectivity in user-item interaction graphs, significantly improving recommendation performance [8], [9].

However, traditional CF models often rely on single-behavior interaction data, which means they only consider user-item interaction data on the target behavior, such as *purchase*. In practical scenarios, user behavior is diverse. For example, on an e-commerce platform, users engage in various actions beyond *purchase*, including *click* and *cart*. The interaction data from these auxiliary behaviors can also provide valuable insights into user preferences. By considering multiple behaviors, we can not only obtain a more comprehensive understanding of user preferences but also alleviate issues such as data sparsity on the target behavior. Recognizing this, researchers have developed Multi-Behavior Recommendation (MBR) approaches [10], [11], [12], [18], [19], [20], [21], [22], [23], [24], [25]. More recently, advanced methods [13], [14], [15], [16], [17] have further pushed the boundaries of exploring complex user-item relationships.

For the sake of subsequent discussion, we refer to the features extracted from the unified heterogeneous graph as *behavior-unified* features, which are dependent on all behaviors and can reflect global user preferences. The features extracted from each behavior are referred to as *behavior-specific* features, which are highly related with the behavior category and reflect local user preferences on different behaviors.

Existing MBR approaches generally fall into two categories. The first category, as shown in Fig. 1(a), treats behaviors independently, constructing distinct GCNs for each behavior to extract behavior-specific features, and then leveraging them for recommendation. Representative methods like MBRCC [13] and MBLFE [14] fall into this group. MBRCC weighs behavior-specific embeddings of users to obtain the final user embeddings and utilizes contrastive clustering learning to optimize user and item representations. MBLFE fuses behavior-specific embeddings based on behavioral interaction data proportions and leverages expert networks to model latent factors for fused embeddings. While effective in capturing behavior-specific signals and

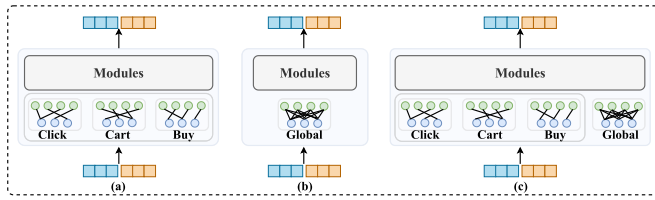


Fig. 1. Multi-behavior recommendation framework. (a) Preference features are mainly from behavior-specific local interaction graphs and involved in the whole process of subsequent modules [10], [11], [12], [13], [14]. (b) Preference features from the behavior-unified global interaction graph are utilized for recommendation or as feature initializer for subsequent modules [15], [16], [17]. (c) Preference features come from the behavior-unified global and behavior-specific local interaction graphs and subsequent modules can leverage and fuse them. This framework is proposed in our work.

modeling behavioral local nuances, they often overlook the unified interaction patterns across behaviors, which reflect the user's inherent global preferences. The second category, as shown in Fig. 1(b), constructs a unified heterogeneous graph to extract behavior-unified features. However, methods like BMGCN [16] use these features directly for recommendations without utilizing behavior-specific preference features, while BCIPM [15] and UIPL [17] use them as feature initialization for subsequent modeling. Specifically, they use initialized features for behavior-contextualized item preference modeling and invariant preference generation, respectively, but behavior-unified patterns are weakened in these subsequent modeling processes. Both categories lack coordinated utilization of unified interaction patterns and behavior-specific preferences, resulting in suboptimal recommendations. To exemplify, consider a user who has recently mostly clicked on high-thermal sleeping bags, added portable grills to the cart, and purchased lightweight trekking poles. Without the behavior-unified underlying preference pattern for a *multi-day wilderness trekking trip*, relying solely on fragmented behavioral local preference such as *sleep comfort*, *outdoor dining*, and *hiking gear*, might lead to unrelated hiking gear recommendation that lacks the global camping context. Conversely, the unified pattern typically oversmooth behavior-specific signals, such as the loss of the critical lightweight requirement indicated by the purchased poles, causing suboptimal recommendation like heavy-duty, non-collapsible tents.

Thus, the current fundamental challenge in the MBR field lies in *how to effectively integrate unified behavior interaction patterns with behavior-specific signals, that is, to achieve the unification of behavior-unified and behavior-specific preferences*. User interaction behavior is driven by both intrinsic global preferences and specific behavioral interests. Integrating these two types of features is not a simple information superposition, but a necessary requirement to bridge the user's unified interests and specific behavioral preferences. However, existing methods fail to align and integrate these two different levels of abstraction, resulting in incomplete and suboptimal representations. To address these challenges, we propose an innovative approach to *fuse behavior-unified and behavior-specific features to unify global and local user preferences for multi-behavior recommendation*. Besides, to promote user preference learning, we propose to apply a preference alignment and refinement process

to the fused features. It is clear that our approach belongs to the framework in Fig. 1(c).

Towards this goal, we propose **Unifying Global and Local Preferences (UGLP)** for multi-behavior recommendation. Unlike previous works, UGLP is designed to explicitly model the interplay between behavior-unified global and behavior-specific local features. Specifically, we design a novel behavior feature fusion network composed of two modules, each serving a distinct purpose. The first module is the global fusion module which performs fusion at the graph-granularity level, enriching each behavior-specific feature with signals from the unified interaction graph. Specifically, each layer performs graph convolution operations to extract both behavior-unified and behavior-specific features based on the features computed in the previous layer. Then, a gating mechanism is employed to dynamically weigh the importance of behavior-unified against behavior-specific features and fuse them to acquire *globally fused features*. This ensures that the representations of each behavior is informed by the user's unified interaction context. The second module is the local fusion module which focuses on the behavior-granularity level and aims to leverage rich behavior-specific information of multiple behaviors to generate comprehensive user representations for the target behavior after the behavior-specific features are enriched by behavior-unified signals. Globally fused features for each behavior are combined through an additional gating network to get *cross-behavior fused features*. This hierarchical design, transitioning from graph-level coordination to behavior-level fusion, ensures that the target behavior representations are enriched by both the global context and auxiliary local signals. Besides, we utilize a contrastive learning module that aligns globally fused features with cross-behavior fused features, promoting preference alignment and knowledge transfer from auxiliary behaviors to the target behavior. Additionally, we employ a GCN-based refinement module to adjust the fused features to enhance the crucial global preference information after the cross-behavior feature fusion process.

In a nutshell, the main contributions are listed as follows:

- We identify that a critical limitation of existing MBR methods lies in their inability to integrate behavior-unified interaction patterns and behavior-specific signals, thereby failing to achieve a true unification of behavior-unified global and behavior-specific local features. Thus, we innovatively propose UGLP to bridge this gap.
- We design a behavior feature fusion network that conducts global and local fusion on behavior-unified and behavior-specific features, providing a hierarchical learning process to model comprehensive user preferences.
- We employ a contrastive learning module to promote preference alignment and knowledge transfer and a GCN refinement module to enhance the crucial global preference information after cross-behavior feature fusion.
- We conduct comprehensive experiments on three real-world datasets, and compare our method with twelve baseline methods. Our approach exhibits significant performance improvement over state-of-the-art methods, verifying practical applicability and robustness across various scenarios.

## II. RELATED WORK

Recently, MBR [20], [24], [26], [27] methods have played a significant role in mitigating data sparsity issues and made remarkable progress. According to whether they are GCN-based or not, MBR methods can be categorized into two main groups: non-GCN-based methods and GCN-based methods. Both of them aim to leverage the diversity of user behaviors and the correlation between different user behaviors.

In the field of non-GCN-based MBR, early methods typically manage multi-behavior data through multiple matrix factorization techniques [18], [19], [28], [29] or by developing new sampling strategies [26], [30], [31]. With advancements in deep learning [32], [33], [34], researchers have begun to develop MBR models based on deep neural networks (DNN), aiming to reveal deeper latent relationships between users and items. DNN-based models often learn embeddings from each behavior and integrate them for prediction on the target behavior. For instance, DIPN [35] and MATN [36] capture the cross-behavior relationships for embedding learning and aggregation by employing various attention mechanisms. In contrast, NMTR [10] utilizes a multi-task learning model, where all user behaviors are prediction targets, and the prediction scores of earlier behaviors are used for subsequent behavior predictions.

Compared to non-GCN-based models, GCN-based methods [11], [21], [37], [38], [39], [40] model collaborative information between users and items by constructing user-item interaction graphs and performing graph convolution operations on these graphs. Additionally, CRGCN [27] and MB-CGCN [41] take into account the cascading relationships between multiple behaviors and achieve great performance by building cascaded graph convolutional networks to capture user preferences. Based on MB-CGCN [41], PKEF [12] further incorporates a parallel knowledge fusion module and feature transformation operations. More recent advancements include the use of contrastive learning [11], [13], [42], [43]. MBSSL [43] leverages a behavior-aware GCN network with self-attention mechanisms to achieve behavior-specific embedding propagation and cross-behavior dependency modeling, and incorporate multi-behavior self-supervised learning to learn the user and item embeddings. MBRCC [13] employs GCNs to obtain the user and item embeddings for different behaviors, then fuses user embeddings and utilizes contrastive clustering learning to improve the quality of the user and item embeddings. Additionally, MBLFE [14] leverages Mixture of Experts (MoE) for multi-behavior recommendation, which first uses GCNs to get the embeddings of users and items for different behaviors, then fuses these embeddings based on behavior interaction data proportions and employs expert networks to model latent factors for fused embeddings. Additionally, some recent approaches focus on the usage of behavior-unified features. BMGCN [16] construct a behavior merging graph based on partial order theory to extract behavior-unified features and then directly utilize them for recommendation. BCIPM [15] uses behavior-unified features from the constructed unified interaction graph to initialize features for subsequent behavior-contextualized item preference modeling and GCN enhancement to enhance user preferences within

the target behavior, while UIPL [17] utilizes behavior-unified initialized features for subsequent environmental representation learning and invariant preference generation.

However, existing approaches fail to integrate behavior-unified interaction patterns and behavior-specific signals in their designs, which means they cannot achieve the unification of behavior-unified global and behavior-specific local preference features. Our method addresses this problem through a behavior feature fusion network that conducts global and local fusion to unify behavior-unified and behavior-specific features, fundamentally demonstrating its design advantages. Furthermore, we incorporate a contrastive learning module and GCN refinement module to further improve the recommendation performance. Comparative experimental results in Section IV validate the effectiveness of the overall design.

## III. METHODOLOGY

### A. Problem Formulation

Let  $\mathcal{U} = \{u_1, \dots, u_M\}$  and  $\mathcal{I} = \{i_1, \dots, i_N\}$  denote the sets of users and items, respectively. We consider  $K$  types of behaviors and behavior  $K$  is the target behavior. The multi-behavior interactions are represented by a set of adjacency matrices  $\mathcal{M} = \{M^1, \dots, M^K\}$ , where  $M^k \in \{0, 1\}^{M \times N}$ . An entry  $m_{ui}^k = 1$  indicates that user  $u$  has interacted with item  $i$  under behavior  $k$ , otherwise  $m_{ui}^k = 0$ . Based on  $\mathcal{M}$ , we derive a behavior-unified interaction matrix  $\mathcal{M}^{uni}$ , representing the global view of user-item connectivity:  $m_{ui}^{uni} = \mathbb{I}(\sum_{k=1}^K m_{ui}^k > 0)$ , where  $\mathbb{I}(\cdot)$  is the indicator function.

Let  $\mathbf{e}_u^{(g)}$  denote the global preference representation learned from  $\mathcal{M}^{uni}$ , and  $\mathbf{e}_u^{(loc,k)}$  denote the local preference representation specific to behavior  $k$  learned from  $M^k$ . Aforementioned existing methods are typically dominated by one type of representation. When behavior-specific features prevail, user preferences are constructed based on  $\text{Agg}(\{\mathbf{e}_u^{(loc,k)}\}_{k=1}^K)$ , which tends to overlook the unified preference patterns  $\mathcal{S} = \mathbf{e}_u^{(g)} \setminus \bigcup_k \mathbf{e}_u^{(loc,k)}$ . Conversely, when behavior-unified representations dominate, this may neglect behavior-specific signals  $\mathcal{D} = \bigcup_k \mathbf{e}_u^{(loc,k)} \setminus \mathbf{e}_u^{(g)}$ , which capture distinct requirements across behaviors.

Therefore, our method formulates the MBR task as learning a dual-fusion mapping  $f_{\Theta} : (\mathcal{M}^{uni}, \mathcal{M}) \rightarrow \hat{y}_{ui}$ , which predicts the probability of user  $u$  interacting with item  $i$  under the target behavior. The core objective is to integrate behavior-unified and behavior-specific information by fusing  $\mathbf{e}_u^{(g)}$  and  $\{\mathbf{e}_u^{(loc,k)}\}_{k=1}^K$  to preserve both the global preference patterns  $\mathcal{S}$  and behavioral local signals  $\mathcal{D}$ .

### B. UGLP

In this section, we focus on the details of the UGLP approach. The architecture of our UGLP is illustrated in Fig. 2. In UGLP, the behavior feature fusion network consists of two parts: the global fusion module and the local fusion module. The global fusion module employs a gate network to fuse behavior-specific and behavior-unified features. Then, the local fusion module

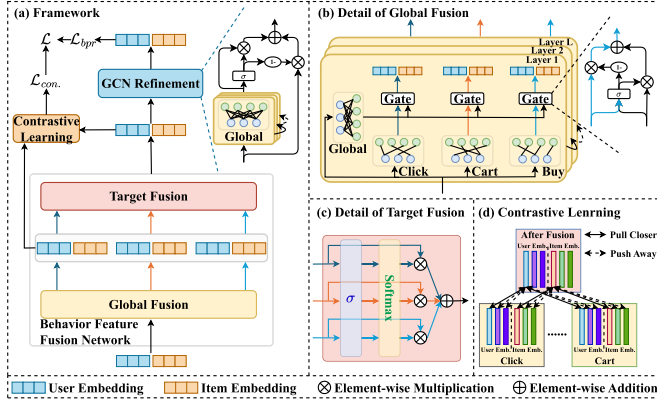


Fig. 2. Overview of our method UGLP. (a) Overall framework of UGLP. The proposed method employs a behavior feature fusion network to fuse behavior-unified global and behavior-specific local preference features into fully fused representations. A contrastive learning module aligns preferences and transfers knowledge across behaviors, while a GCN refinement module further enhances the critical global preference information after fusion. (b) Detail of global fusion. Each layer in this module performs graph convolution operations to extract behavior-unified and behavior-specific features based on the previous layer's outputs. These features are then scaled and integrated via an efficient gating network. (c) Detail of local fusion. In this module, the global fusion features of each behavior are passed are fed into an additional gating network to obtain gating weights. A softmax operation normalizes these weights, which are then used to compute a weighted fusion of the global fusion features, resulting in enhanced representations. (d) Detail of contrastive learning module. This module utilizes contrastive learning to promote preference alignment and knowledge transfer.

utilizes another gate network to perform cross-behavior fusion on the fused features obtained from the global fusion module. This process aims to capture comprehensive and fine-grained user preferences. The contrastive learning module plays a role in promoting alignment between fused and behavior-specific preferences and facilitating preference knowledge transfer from auxiliary behaviors to the target behavior. The GCN refinement module is responsible for refining the fused features, ensuring that both global and local user preference information are preserved. Our approach leverages the contrastive loss and BPR loss [4] for joint optimization.

1) *Embedding Layer*: In industrial contexts, both users and items are typically represented by high-dimensional one-hot vectors. Adopting the prevalent methodology in current recommendation systems [8], [10], [21], we link each user and item to an ID embedding. More concretely, let  $\mathbf{E}_u \in \mathbb{R}^{M \times d}$  and  $\mathbf{E}_i \in \mathbb{R}^{N \times d}$  represent the embedding matrices for users and items respectively, where  $M$  indicates the number of users,  $N$  denotes the number of items and  $d$  represents the size of the embeddings. For a given user-item pair  $(u, i)$ , the embedding lookup operation is employed to retrieve the corresponding embedding vectors for user  $u$  and item  $i$ :

$$\mathbf{e}_u^0 = \mathbf{E}_u \cdot \mathbf{P}_u, \quad \mathbf{e}_i^0 = \mathbf{E}_i \cdot \mathbf{Q}_i, \quad (1)$$

where  $\mathbf{P}_u \in \mathbb{R}^M$  and  $\mathbf{Q}_i \in \mathbb{R}^N$  represent user  $u$ 's and item  $i$ 's one-hot IDs respectively.

2) *Behavior Feature Fusion Network*: Inspired by the message-passing architecture in GCNs, we develop a behavior feature fusion network based on GCN to capture complex CF

signals between nodes and among behaviors. This network aims to fully fuse behavior-unified global and behavior-specific local features, consisting of global and local fusion modules. In this network, the global fusion module incorporates global user preference features into local user preference features by integrating behavior-unified features and behavior-specific features via a gating network. Then, the local fusion module further leverages rich information from the single behavior feedback by combining globally fused features for each behavior with those from other behaviors through an additional gating network.

*Global Fusion Module*: As mentioned above, behavior-unified features contains rich global user preference information while behavior-specific features are associated with local user preferences for specific behaviors. To incorporate global user preferences into local user preferences to enrich specific behaviors' preference information, we design the global fusion module based on LightGCN. Since user preferences evolve through different orders of neighborhood interactions, the global fusion module adopts a hierarchical fusion approach to capture the multi-level synergy between behavior-unified and behavior-specific features. It fuses behavior-unified and behavior-specific features via a gating network at each layer, and subsequently passes the fusion results to the subsequent layer to ensure that the extracted global insights can guide the local preference learning at every abstraction level.

More concretely, in the global fusion module, we obtain behavior-unified features by performing graph convolution operations on a unified user-item interaction graph as:

$$\begin{aligned} \mathbf{e}_u^{(g,l)} &= \sum_{i \in N_u^g} \frac{1}{\sqrt{|N_u^g|} \sqrt{|N_i^g|}} \mathbf{e}_i^{(g,l-1)}, \\ \mathbf{e}_i^{(g,l)} &= \sum_{u \in N_i^g} \frac{1}{\sqrt{|N_i^g|} \sqrt{|N_u^g|}} \mathbf{e}_u^{(g,l-1)}, \end{aligned} \quad (2)$$

where  $\mathbf{e}_u^{(g,l)}$  and  $\mathbf{e}_i^{(g,l)}$  are the updated embeddings of user  $u$  and item  $i$ , obtained from the  $l$ -th layer on the unified interaction graph.  $N_u^g$  represents the set of items that have been interacted with by user  $u$  in the unified interaction graph, and  $N_i^g$  refers to the set of users that have interacted with item  $i$  in the unified interaction graph. This process enables the extraction of generalized features that are dependent on all user behaviors, thereby capturing the overall patterns and preferences in all user-item interactions. It is crucial to highlight that  $\mathbf{e}_u^{(g,0)}$  and  $\mathbf{e}_i^{(g,0)}$  are the initialized embeddings for user  $u$  and item  $i$ , respectively. Analogously, behavior-specific features can be obtained by performing graph convolution operations on user-item interaction graphs specific to each behavior as:

$$\begin{aligned} \mathbf{e}_u^{(k,l)} &= \sum_{i \in N_u^k} \frac{1}{\sqrt{|N_u^k|} \sqrt{|N_i^k|}} \mathbf{e}_i^{(k,l-1)}, \\ \mathbf{e}_i^{(k,l)} &= \sum_{u \in N_i^k} \frac{1}{\sqrt{|N_i^k|} \sqrt{|N_u^k|}} \mathbf{e}_u^{(k,l-1)}, \end{aligned} \quad (3)$$

where  $\mathbf{e}_u^{(k,l)}$  and  $\mathbf{e}_i^{(k,l)}$  are the updated embeddings of user  $u$  and item  $i$ , obtained from the  $l$ -th layer on the interaction

graph of behavior  $k$ .  $N_u^k$  represents the set of items that have been interacted with by user  $u$  in the behavior  $k$ 's interaction graph, and  $N_i^k$  refers to the set of users that have interacted with item  $i$  in the behavior  $k$ 's interaction graph. This process allows for the extraction of detailed features that are closely tied to individual user behaviors, thereby capturing the unique user preferences associated with each specific behavior. It is worth noting that when computing behavior-specific features, we integrate them with behavior-unified features from the same layer, which means in (3), when  $l > 1$ ,  $\mathbf{e}_u^{(k,l-1)}$  represents the embeddings of user  $u$  after fusing  $\mathbf{e}_u^{(g,l-1)}$  and  $\mathbf{e}_u^{(k,l-1)}$  in the  $(l-1)$ -th layer while  $\mathbf{e}_u^{(k,0)}$  refers to the initial embeddings of user  $u$ . The same principle applies to  $\mathbf{e}_i^{(k,l-1)}$  for item  $i$ . Next, we will discuss how to fuse behavior-unified features  $\mathbf{e}_u^{(g,l)}$  and behavior-specific features  $\mathbf{e}_u^{(k,l)}$  in the  $l$ -th layer.

In order to effectively fuse behavior-unified features  $\mathbf{e}_u^{(g,l)}$  and behavior-specific features  $\mathbf{e}_u^{(k,l)}$ , we use a simple but effective gating network that can dynamically adjust the weights of behavior-unified global and behavior-specific local features. For each behavior  $k$  in the  $l$ -th layer, the corresponding fusing weights for user  $u$   $G_u^{(k,l)}$  and item  $i$   $G_i^{(k,l)}$  are computed as follows:

$$\begin{aligned} G_u^{(k,l)} &= \sigma \left( \mathbf{e}_u^{(g,l)} \mathbf{W}_{G_{k_1}} + \mathbf{e}_u^{(k,l)} \mathbf{W}_{G_{k_2}} + b_{G_k} \right), \\ G_i^{(k,l)} &= \sigma \left( \mathbf{e}_i^{(g,l)} \mathbf{W}_{G_{k_1}} + \mathbf{e}_i^{(k,l)} \mathbf{W}_{G_{k_2}} + b_{G_k} \right), \end{aligned} \quad (4)$$

where  $\mathbf{W}_{G_{k_1}} \in \mathbb{R}^{d \times d}$ ,  $\mathbf{W}_{G_{k_2}} \in \mathbb{R}^{d \times d}$  and  $b_{G_k} \in \mathbb{R}^d$ ; and  $\sigma(\cdot)$  is the sigmoid function. The rationale behind this design is to enable a mutual interaction between two types of information sources. By simultaneously considering both behavior-unified and behavior-specific inputs, the gating network acts as an interactive mediator that evaluates the relative importance of them, ensuring a balanced fusion that enhances specific behavioral representations.

With the fusing weights  $G_u^{(k,l)}$  and  $G_i^{(k,l)}$ , the globally fused features  $\mathbf{e}_u^{(k,l)}$  and  $\mathbf{e}_i^{(k,l)}$  can be acquired by the weighted sum of behavior-unified features  $\mathbf{e}_u^{(g,l)}$  and behavior-specific features  $\mathbf{e}_u^{(k,l)}$  as follows:

$$\begin{aligned} \mathbf{e}_u^{(k,l)} &= G_u^{(k,l)} \otimes \mathbf{e}_u^{(g,l)} + \left(1 - G_u^{(k,l)}\right) \otimes \mathbf{e}_u^{(k,l)}, \\ \mathbf{e}_i^{(k,l)} &= G_i^{(k,l)} \otimes \mathbf{e}_i^{(g,l)} + \left(1 - G_i^{(k,l)}\right) \otimes \mathbf{e}_i^{(k,l)}, \end{aligned} \quad (5)$$

where  $\otimes$  means the element-wise product.

After the completion of  $L$  layers of graph convolution, in the same way as LightGCN, this network further integrates the embeddings obtained at each layer to construct the representations of a user and an item under the behavior  $k$  as follows:

$$\mathbf{e}_u^{(k)} = \sum_{l=0}^L \alpha_l \mathbf{e}_u^{(k,l)}, \quad \mathbf{e}_i^{(k)} = \sum_{l=0}^L \alpha_l \mathbf{e}_i^{(k,l)}, \quad (6)$$

where  $\alpha_l$  represents the weighting factor for the fused embeddings from each layer, which we set uniformly as  $\frac{1}{L+1}$  in our experiments following the setting of LightGCN, and  $\mathbf{e}_u^{(k,l)}$  and  $\mathbf{e}_i^{(k,l)}$  are the fused embeddings of user  $u$  and item  $i$  under

the behavior  $k$  at layer  $l$ , respectively. This integration process captures the multi-layer information to form a comprehensive representation.

**Local Fusion Module:** After fusing behavior-specific and behavior-unified features using the global fusion module, we obtain globally fused features, denoted as  $\mathbf{e}_u^{(k)}$  for user  $u$  and  $\mathbf{e}_i^{(k)}$  for item  $i$ , specific to behavior  $k$ . However, these globally fused features still contain rich information from the single behavior  $k$  feedback. Since different behaviors exhibit varying degrees of correlation with the target intent, we utilize another gating network to adaptively quantify the contribution of each individual behavior's globally fused feature to the cross-behavior fusion representations. By employing this additional gating network, the local fusion module supports a flexible mechanism to highlight the most informative behavior features while suppressing irrelevant ones, allowing for further improvement of the target behavior's representations.

We denote  $C_u^k$  and  $C_i^k$  as the fusing gate values under behavior  $k$  for the user  $u$  and item  $i$ , respectively. Both  $C_u^k$  and  $C_i^k$  can be formulated as follows:

$$C_u^k = \sigma \left( \mathbf{e}_u^{(k)} \mathbf{W}_C + b_C \right), \quad C_i^k = \sigma \left( \mathbf{e}_i^{(k)} \mathbf{W}_C + b_C \right), \quad (7)$$

Where  $\mathbf{W}_C \in \mathbb{R}^{d \times d}$ ,  $b_C \in \mathbb{R}^d$ , and  $\sigma(\cdot)$  means the sigmoid function that can restrict  $C_u^k$  and  $C_i^k$  within the interval  $(0,1)$ .

Based on the fusing gate values  $C_u^k$  and  $C_i^k$ , we further use the softmax function to determine the fusion contribution of each behavior as follows:

$$C_u^{(K)} = \frac{e^{C_u^K}}{\sum_{k'=1}^K e^{C_u^{k'}}}, \quad C_i^{(K)} = \frac{e^{C_i^K}}{\sum_{k'=1}^K e^{C_i^{k'}}}, \quad (8)$$

With the above fusing weights, the cross-behavior fusion features  $\mathbf{e}_u^{(K)}$  and  $\mathbf{e}_i^{(K)}$  for the target behavior  $K$  can be acquired by the weighted sum of the globally fused features  $\mathbf{e}_u^{(k')}$  and  $\mathbf{e}_i^{(k')}$  respectively as:

$$\mathbf{e}_u^{(K)} = \sum_{k'=1}^K C_u^{k'} \otimes \mathbf{e}_u^{(k')}, \quad \mathbf{e}_i^{(K)} = \sum_{k'=1}^K C_i^{k'} \otimes \mathbf{e}_i^{(k')}, \quad (9)$$

where  $\otimes$  represents the element-wise product. It is worth noting that the cross-behavior fusion features  $\mathbf{e}_u^{(K)}$  and  $\mathbf{e}_i^{(K)}$  are obtained by fusing globally fused features across different behaviors, which enables fused features to leverage information from different single behavior feedbacks to enhance the learning of representations for the target behavior.

**3) Contrastive Learning Module:** To promote the alignment between fused and behavior-specific preferences after the feature fusion process and facilitate knowledge transfer from auxiliary behaviors to the target behavior, we employ a contrastive learning module between cross-behavior fusion features for the target behavior, denoted as  $\mathbf{e}_u^{(K)}$  and  $\mathbf{e}_i^{(K)}$ , and globally fused features of the all behaviors, denoted as  $\mathbf{e}_u^{(k)}$  and  $\mathbf{e}_i^{(k)}$  (for  $k = 1, \dots, K$ ). The contrastive learning module can help to refine cross-behavior fusion features for the target behavior and align them with those for the auxiliary behaviors, reducing noise and promoting knowledge transfer, ultimately enhancing the quality of the target behavior's features.

Specifically, cross-behavior fusion features for the target behavior  $K$  are to be contrasted with globally fused features for each behavior  $k$  from  $\{1, 2, \dots, K\}$ . Following the common practice, we treat the views of the same node as positive pairs, namely  $\{(\mathbf{e}_u^{(K)}, \mathbf{e}'_u^{(k)}) | u \in \mathcal{U}\}$ , and the views of any different nodes as the negative pairs, namely  $\{(\mathbf{e}_u^{(K)}, \mathbf{e}'_v^{(k)}) | u, v \in \mathcal{U}, u \neq v\}$ . The contrastive loss between cross-behavior fusion features for the target behavior  $K$  and globally fused features for the behavior  $k$  is computed as:

$$\mathcal{L}_{con.,user}^{K,k} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} -\log \frac{\exp(\phi(\mathbf{e}_u^{(K)}, \mathbf{e}'_u^{(k)})/\tau)}{\sum_{v \in \mathcal{U}} \exp(\phi(\mathbf{e}_u^{(K)}, \mathbf{e}'_v^{(k)})/\tau)}, \quad (10)$$

where  $\phi(\cdot)$  represents the inner-product between two vectors and  $\tau$  denotes the temperature hyper-parameter in the softmax function. With the assistance of the contrastive loss, the distances between the features of the same nodes on the target behavior and auxiliary behaviors are reduced, while the distances between features of different nodes are increased. By minimizing the distance between the features of the same nodes under different behaviors, we encourage similar representations for these nodes, which facilitates preference alignment and knowledge transfer across behaviors. At the same time, by maximizing the distance between features of different nodes, we ensure that the representations of different nodes remain distinct and capture their specific characteristics.

Analogously integrated with the item side's contrastive loss, the contrastive loss is  $\mathcal{L}_{con.}^{K,k} = \mathcal{L}_{con.,user}^{K,k} + \mathcal{L}_{con.,item}^{K,k}$ . Thus, the final contrastive loss can be calculated as  $\mathcal{L}_{con.} = \sum_{k=1}^K \mathcal{L}_{con.}^{K,k}$ .

4) *GCN Refinement Module*: Additionally, to enhance the crucial behavior-unified global preference information after the behavior feature fusion process, inspired by [15], we introduce a GCN Refinement module to refine the behavior-unified global preferences of the target behavior features, enhancing the accuracy and effectiveness of our multi-behavior recommendation approach. For simplicity, we directly utilize LightGCN as the GCN refinement module. The module executes graph convolution operations on the unified interaction graph. Thereby, the embeddings of a user  $u$  and an item  $i$  in the  $l$ -th layer under the target behavior  $K$  can be obtained as follows:

$$\begin{aligned} \hat{\mathbf{e}}_u^{(l)} &= \sum_{i \in N_u^g} \frac{1}{\sqrt{|N_u^g|} \sqrt{|N_i^g|}} \hat{\mathbf{e}}_i^{(l-1)}, \\ \hat{\mathbf{e}}_i^{(l)} &= \sum_{u \in N_i^g} \frac{1}{\sqrt{|N_i^g|} \sqrt{|N_u^g|}} \hat{\mathbf{e}}_u^{(l-1)}, \end{aligned} \quad (11)$$

where  $\hat{\mathbf{e}}_u^{(0)}$  and  $\hat{\mathbf{e}}_i^{(0)}$  represents the embeddings generated by the behavior feature fusion network for the user  $u$  and the item  $i$  under the target behavior  $K$ , namely  $\hat{\mathbf{e}}_u^{(0)} = \mathbf{e}_u^{(K)}$  and  $\hat{\mathbf{e}}_i^{(0)} = \mathbf{e}_i^{(K)}$ . After propagating through  $L$  layers, to obtain the global preference refinement embeddings for user  $u$  and item  $i$ , we sum the embeddings from each layer and take the average as:

$$\hat{\mathbf{e}}_u^{(K)} = \frac{1}{L+1} \sum_{l=0}^L \hat{\mathbf{e}}_u^{(l)}; \quad \hat{\mathbf{e}}_i^{(K)} = \frac{1}{L+1} \sum_{l=0}^L \hat{\mathbf{e}}_i^{(l)}, \quad (12)$$

To evaluate the inherent reliability and significance of the newly refined signals independently, we leverage another network to compute the refinement fusion weights as (13), allowing the model to adaptively decide how much this final unified update should impact the final embeddings.

$$R_u^{(K)} = \sigma(\hat{\mathbf{e}}_u^{(K)} \mathbf{W}_R + b_R), \quad R_i^{(K)} = \sigma(\hat{\mathbf{e}}_i^{(K)} \mathbf{W}_R + b_R), \quad (13)$$

where  $\mathbf{W}_R \in \mathbb{R}^{d \times d}$ ,  $b_R \in \mathbb{R}^d$ , and  $\sigma(\cdot)$  is the sigmoid function.

With the fusing weights  $R_u^{(K)}$  and  $R_i^{(K)}$ , the final embeddings can be acquired by the weighted sum of global preference refinement features and cross-behavior fusion features as follows:

$$\begin{aligned} \tilde{\mathbf{e}}_u &= R_u^{(K)} \otimes \hat{\mathbf{e}}_u^{(K)} + (1 - R_u^{(K)}) \otimes \mathbf{e}_u^{(K)}, \\ \tilde{\mathbf{e}}_i &= R_i^{(K)} \otimes \hat{\mathbf{e}}_i^{(K)} + (1 - R_i^{(K)}) \otimes \mathbf{e}_i^{(K)}, \end{aligned} \quad (14)$$

where  $\otimes$  means the element-wise product;  $\tilde{\mathbf{e}}_u$  and  $\tilde{\mathbf{e}}_i$  are the final representations of user  $u$  and item  $i$  for the target behavior  $K$ , respectively.

### C. Overall Optimization

Our objective is to optimize the model for top- $n$  recommendation which means recommending a set of  $n$  items to a user according to their predicted preference scores. Specifically, our model calculates the inner product of the final representations of user  $u$  and item  $i$  for the target behavior  $K$  to acquire the preference score as  $\hat{y}_{ui} = \tilde{\mathbf{e}}_u \tilde{\mathbf{e}}_i^T$ . The recommender system will recommend items to the user according to their scores in descending order.

To achieve optimization objective, the pairwise Bayesian personalized ranking loss (BPR) function [4] is adopted to help the model training. The BPR function can ensure that the user  $u$ 's score for the positive item  $i^+$  is higher than for the negative item  $i^-$ , where  $i^+$  represents the item the user has interacted with and  $i^-$  indicates the item the user has not interacted with on the target behavior  $K$ . The BPR loss function in our model is calculated as:

$$\mathcal{L}_{bpr} = \sum_{(u, i^+, i^-) \in O} -\ln \sigma(\hat{y}_{ui^+} - \hat{y}_{ui^-}), \quad (15)$$

where  $O = \{(u, i^+, i^-) \mid (u, i^+) \in R^+, (u, i^-) \in R^-\}$  and  $R^+$  represents sets of items interacted with by user  $u$  and  $R^-$  indicates sets of items not interacted with by user  $u$  on the target behavior  $K$ .  $\sigma(\cdot)$  denotes the sigmoid function.

Eventually, we combine the contrastive loss  $\mathcal{L}_{con.}$  and the BPR loss  $\mathcal{L}_{bpr}$  to obtain the overall optimization as follows:

$$\mathcal{L} = \lambda \mathcal{L}_{con.} + (1 - \lambda) \mathcal{L}_{bpr} + \beta \|\Theta\|_2, \quad (16)$$

where  $\lambda$  is a hyper-parameter that controls the relative weights of  $\mathcal{L}_{con.}$  and  $\mathcal{L}_{bpr}$ ,  $\Theta$  denotes all trainable parameters in our model and  $\|\cdot\|_2$  is the  $L_2$  regularization which avoids the risk of overfitting and is controlled by another hyper-parameter  $\beta$ .

#### D. Time Complexity Analysis

The overall computational complexity of our model is dominated by three components: behavior feature fusion network, contrastive learning module and GCN refinement module. In the behavior feature fusion network, the global fusion module, which includes representation extraction and hierarchical fusion, has a time complexity of  $O((K+1)LEd + KLNd^2)$ , where  $K$  is the number of behaviors,  $L$  is the number of GCN layers,  $E$  represents the number of edges,  $N$  denotes the number of nodes, and  $d$  is the embedding size, while the local fusion module leads to a time complexity of  $O(KNd^2)$ . The contrastive learning module utilizing in-batch negatives contributes a time complexity of  $O(KNBd)$ , where  $B$  denotes the batch size. The global refinement module has a time complexity of  $O(LEd + Nd^2)$ . Summing the dominant terms, the overall time complexity of one training epoch can be approximated as:  $O((K+2)LEd + (KL + K + 1)Nd^2 + KNBd)$ . In practice,  $L$  is a small constant,  $d$  is typically not large, and the number of behaviors  $K$  is typically limited. Therefore, the overall complexity remains acceptable for various datasets, as confirmed by our efficiency analysis in Section IV-G.

### IV. EXPERIMENTS

#### A. Experiment Settings

1) *Dataset*: Our experiments have been conducted on three public real-world datasets: Taobao,<sup>1</sup> Tmall<sup>2</sup> and Yelp.<sup>3</sup> The brief description of the three datasets are as follows:

- *Taobao*: This is an open dataset acquired from Taobao, which contains about 2.3 million interactions from 48,749 users and 39,493 items, and involves three types of interaction behavior, i.e., *click*, *cart*, and *purchase*. Among these interaction behaviors, *purchase* is regarded as the target behavior and contains around 259,771 interactions.
- *Tmall*: This dataset is sourced from Tmall. It contains about 2.3 million interactions between 41,738 users and 11,953 items, with four types of interaction behavior, i.e., *click*, *collect*, *cart*, and *purchase*, where the target behavior *purchase* encompasses around 287,158 interactions.
- *Yelp*: This dataset is collected from Yelp, which contains *tips* and *ratings* which range from 1(worst) to 5(best) stars with 0.5 star as increment. Building upon prior research [15], [36], [37], *ratings* can be classified into three distinct behaviors: *dislike* (*ratings*  $\in [0, 2]$ ), *neutral* (*ratings*  $\in (2, 4)$ ), and *like* (*ratings*  $\in [4, 5]$ ), among which 677,343 *like* interactions of all 1.4 million interactions are considered as the target behavior, with *tips*, *dislike*, and *neutral* regarded as auxiliary behaviors.

For the three datasets, to process duplicated user-item interactions, we adhere to the approaches of previous works [10], [21] to keep only the earliest occurrence of each interaction.

2) *Evaluation Protocols*: Following the previous studies [10], [21], [44], we employ the widely used leave-one-out strategy during our evaluation process. In this strategy, for each

user, we designate the most recent interacted item as well as all non-interacted items as the test set, while utilizing all other interacted items for training purposes. During the evaluation phase, we rank all items in the test set based on the scores predicted by the recommendation model. Subsequently, we select the top- $n$  items to assess the performance of our model and baseline models. To evaluate the effectiveness of the top- $n$  recommendations, we adopt two popular metrics Hit Ratio (HR@ $n$  or H@ $n$ ) and Normalized Discounted Cumulative Gain (NDCG@ $n$  or N@10).

3) *Baselines*: To verify the effectiveness of our model, we compare it with several competitive recommendation models, which can be categorized into two groups: (A) Single-behavior models (MF-BPR [4], LightGCN [8]); (B) Multi-behavior models (NMTR [10], MBGCN [21], S-MBRec [11], CRGCN [27], MB-CGCN [41], PKEF [12], BCIPM [15], MBRCC [13], UIPL [17], MBLFE [14]). We briefly describe these models as follows.

##### Single-behavior models:

- *MF-BPR* [4]: This method utilizes matrix decomposition techniques and Bayesian Personalized Ranking Loss Optimization for the top- $n$  recommendation.
- *LightGCN* [8]: This method reduces model complexity by simplifying GCN operation and significantly improves performance.

##### Multi-behavior models:

- *NMTR* [10]: This method predicts each behavior's interaction score which then is transferred to the next behavior, with multi-task learning as the optimization strategy of the model parameters.
- *MBGCN* [21]: This model employs user preference learning on behavior-specific graphs and considers varying importance of multiple behaviors on the target behavior while exploiting the high-order item-item relations.
- *S-MBRec* [11]: This model utilize a star-style contrastive learning strategy to construct a contrastive view pair for the target and each auxiliary behavior.
- *CRGCN* [27]: This method transfers features learned from previous behaviors to the current behavior via a cascading residual network.
- *MB-CGCN* [41]: This model simplifies CRGCN by leveraging LightGCN directly to capture behavioral dependencies in behavioral chains for embedding learning.
- *PKEF* [12]: This model incorporates a parallel knowledge fusion module with a projection mechanism on decoupled multi-expert information to address the challenge of imbalanced data distribution.
- *BCIPM* [15]: This approach mainly devises a behavior-contextualized item preference network to extract item-specific preferences for various behaviors from the constructed unified user-item bipartite graph.
- *MBRCC* [13]: This model weighs behavior-specific embeddings to obtain the final embeddings and utilizes contrastive clustering learning to improve the quality of the embeddings of users and items.
- *UIPL* [17]: This method utilizes behavior-unified features for subsequent environmental representation learning and invariant preference generation.

<sup>1</sup><https://tianchi.aliyun.com/dataset/649>

<sup>2</sup><https://tianchi.aliyun.com/dataset/140281>

<sup>3</sup><https://www.kaggle.com/yelp-dataset/yelp-dataset>

TABLE I

OVERALL PERFORMANCE COMPARISONS ON TAOBAO, TMALL AND YELP (THE BOLD FONT INDICATES THE BEST RESULTS WHILE THE UNDERLINE REPRESENTS THE SECOND BEST RESULTS, AND "REL IMPR." REPRESENTS THE RELATIVE PERFORMANCE IMPROVEMENT COMPARED TO THE SECOND BEST RESULTS). SIGNIFICANCE LEVEL P-VALUE < 0.05

| Method       | Taobao        |               | Tmall         |               | Yelp          |               |
|--------------|---------------|---------------|---------------|---------------|---------------|---------------|
|              | H@10          | N@10          | H@10          | N@10          | H@10          | N@10          |
| MF-BPR [4]   | 0.0178        | 0.0101        | 0.0230        | 0.0124        | 0.0327        | 0.0159        |
| LightGCN [8] | 0.0254        | 0.0138        | 0.0393        | 0.0209        | 0.0400        | 0.0202        |
| NMTR [10]    | 0.0409        | 0.0212        | 0.0517        | 0.0250        | 0.0324        | 0.0160        |
| MBGCN [21]   | 0.0434        | 0.0259        | 0.0549        | 0.0285        | 0.0356        | 0.0183        |
| S-MBRec [11] | 0.0571        | 0.0331        | 0.0694        | 0.0362        | 0.0353        | 0.0173        |
| CRGCN [27]   | 0.1152        | 0.0629        | 0.0840        | 0.0442        | 0.0367        | 0.0178        |
| MB-CGCN [41] | 0.0989        | 0.0470        | 0.1073        | 0.0416        | 0.0355        | 0.0164        |
| PKEF [12]    | 0.1097        | 0.0627        | 0.1118        | 0.0630        | 0.0423        | 0.0210        |
| BCIPM [15]   | 0.1292        | 0.0716        | 0.1414        | 0.0741        | 0.0502        | 0.0244        |
| MBRCC [13]   | 0.1117        | 0.0667        | 0.1315        | 0.0704        | 0.0485        | 0.0224        |
| UIPL [17]    | 0.1328        | 0.0782        | 0.1434        | 0.0790        | 0.0522        | 0.0254        |
| MBLFE [14]   | 0.1343        | 0.0795        | 0.1452        | 0.0787        | 0.0514        | 0.0255        |
| Ours         | <b>0.1560</b> | <b>0.0889</b> | <b>0.1627</b> | <b>0.0882</b> | <b>0.0632</b> | <b>0.0305</b> |
| Rel Impr.    | 16.16%        | 11.82%        | 12.05%        | 11.65%        | 21.07%        | 19.60%        |

- **MBLFE [14]:** This model fuses behavior-specific embeddings based on behavioral interaction data proportions and leverages expert networks to model latent factors for fused embeddings.

4) *Hyper-Parameter Settings:* Our model is implemented utilizing Pytorch. In all experiments, we uniformly set the embedding size to 64 and fix the mini-batch size at 1024. Our model parameters are initialized leveraging the Xavier method. For all GCN modules, the number of GCN layers is fixed at 2. For all the approaches involving pair-wise learning loss [10], four negative samples are selected at random for each positive sample [7], [10]. In our model, the Adam optimizer is used for optimization during the training phase, in which we set the learning rate to 0.0001. Besides, we search  $\lambda$  in [0.1, 0.9] for Taobao and Tmall datasets while search  $\lambda$  in [0.01, 0.09] for Yelp dataset, and set  $\beta$  to 0.0001 in overall optimization. For the hyper-parameters in other benchmark models, according to their original papers, we carefully tune them to ensure a fair comparison.

## B. Overall Performance

The results of the experimental comparison on the Taobao, Tmall and Yelp datasets are presented in Table I. In conclusion, we draw the following findings:

*Effectiveness of UGLP:* Our model consistently achieves the best performance across all three datasets over all baseline models. Concretely, our model outperforms the optimal baselines 16.16% and 11.82% on Taobao, 12.05% and 11.65% on Tmall and 21.07% and 19.60% on Yelp in terms of both recommender system evaluation metrics HR@10 and NDCG@10, respectively. This is a remarkable improvement in recommendation accuracy, particularly when compared to the advanced baselines, UIPL and MBLFE. This substantial progress underscores the effectiveness of our model. Specifically, on the one hand, compared to the first category of SOTA methods like MBRCC and MBLFE, which treat behaviors independently and fuse them for

subsequent modeling where MBRCC leverages contrastive clustering learning and MBLFE utilizes expert networks to model latent factors, our model avoids local suboptimal by incorporating a global behavior-unified context. On the other hand, compared to the second category like BCIPM and UIPL, which typically use behavior-unified features only as feature initialization, our model employs a hierarchical fusion mechanism. This allows the behavior-unified and behavior-specific information to interact dynamically throughout each layer, ensuring that the global preference signal actively guides the learning of local behavioral nuances. Furthermore, by leveraging GCN refinement module based on the unified graph and contrastive learning module, the quality of cross-behavior fusion representations is further enhanced.

*Comprehensive Analysis:* (1) For single-behavior recommendation models, LightGCN outperforms MF-BPR on all three datasets, demonstrating the advantages of GCN models in utilizing the information of higher-order neighbors in the learning of user and item embeddings. (2) For multi-behavior recommendation methods, on Taobao and Tmall datasets, multi-behavior approaches from NMTR to MBLFE tend to outperform single-behavior approaches. This suggests that integrating multiple behaviors can help preference modeling and improve the recommendation performance. However, on the Yelp dataset, multi-behavior approaches from NMTR to MB-CGCN do not outperform LightGCN, which is attributed to the fact that multiple behaviors in Yelp do not have strong correlations as in Taobao and Tmall, and thus the direct fusion of behavior-specific features introduces preference noise and affects the performance. PKEF outperforms LightGCN by optimizing the fusion strategy, BCIPM achieves significant improvement by using behavior-unified features to reduce the noise of fusion, and MBRCC leverages contrastive clustering learning to implicitly utilize knowledge of auxiliary behaviors, while UIPL and MBLFE disentangle important features instead of direct fusion. (3) A key observation is that our model achieves the most significant improvement on Yelp, despite its weaker behavioral correlations compared to Taobao or Tmall. This phenomenon is closely tied to the role of behavior-unified signals. The behavior-unified signals, which are rooted in the unified graph that aggregates all interactions, provide stable and reliable representations of a user's intrinsic preference. Our model's behavior-unified refinement module leverages these stable unified signals to refine the fused representations. In datasets like Yelp, where individual behaviors are loosely coupled and behavior-specific features often contain high variance or inconsistent signals, the quality of fused representations tends to be poorer. In this scenario, the corrective role of behavior-unified signals within the refinement module becomes more pronounced on the Yelp dataset. The results in the Fig. 3 also confirm that behavior-unified signals within the refinement module plays a more significant role on the Yelp dataset. The significant gain on Yelp underscores that explicitly utilizing behavior-unified signals as a refinement guide is especially effective when specific behavioral correlations are weak, thereby proving the strong generalizability of our architecture across different recommendation environments.

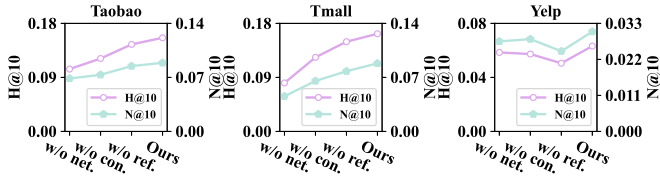


Fig. 3. Ablation study of our proposed method (“w/o. net.,” “w/o. con.” and “w/o. ref.” indicate methods that without behavior feature fusion network, without contrastive learning module and without GCN refinement module, respectively).

### C. Ablation Study

In this section, we conduct comprehensive ablation experiments to assess the effectiveness of key components for the final performance.

1) *Component Effectiveness Analysis*: To investigate component effectiveness, we conduct ablation experiments on the individual design elements to observe the specific influence of the different components. To this end, we compare our model with the following three variants:

- *w/o. net.* : In this variant, we remove the behavior feature fusion network and utilize the user and item embeddings learned from the target behavior interaction data using LightGCN as the input of the downstream modules.
- *w/o. con.* : In this variant, we exclude the contrastive loss and rely primarily on the BPR loss for optimization.
- *w/o. ref.* : In this variant, we remove the GCN refinement module and directly leverage the user and item embeddings from behavior feature fusion network for training and prediction.

The experimental results are reported in Fig. 3.

A comparison of the performance of our model with the other three variants reveals that each variant leads to a distinct level of performance degradation when any key component is removed. The variant without fusion modules exhibits a decline in performance, suggesting that fusing behavior-unified global and behavior-specific local features is a viable and effective strategy. Similarly, the variant devoid of the contrastive learning module demonstrates a substantial reduction in performance, indicating that the fused features necessitate further preference alignment and knowledge transfer. Furthermore, the variant with the GCN refinement module removed also experiences a decline in performance, indicating the necessity of further refinement for the fused features with the constraint of contrastive loss. In conclusion, the ablation experimental outcomes demonstrate that the removal of any component from the model results in a performance decline. These results confirm the efficacy and significant contributions of each component in the model design.

2) *Behavior Feature Fusion Network Analysis*: To validate the effectiveness of the behavior feature fusion network, we conduct ablation experiments on its components to observe the specific impact of different parts. For this purpose, we compare our model with three variants, focusing on the contributions of the global and local fusion modules. Specifically, the three variants we used are as follows: (1) **w/o. glo.&loc.** : removing the global and local fusion modules simultaneously; (2) **w/o. glo.** : removing only the global fusion module; and (3) **w/o. loc.** : removing only the local fusion module. The experimental

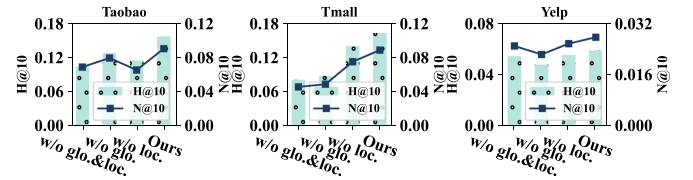


Fig. 4. Behavior feature fusion network analysis (“w/o. glo.&loc.” represents the network without both global fusion module and local fusion module, “w/o. glo.” means the network without global fusion module and “w/o. loc.” indicates the network without local fusion module).

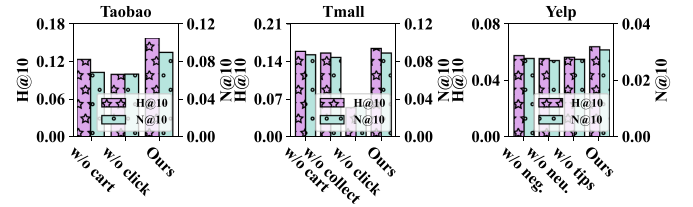


Fig. 5. Auxiliary behavior effectiveness analysis.

**loc.** : removing only the local fusion module. The experimental results are exhibited in Fig. 4. According to Fig. 4, we can clearly observe that for Taobao and Tmall, which have sparse interactions and strong behavior correlations, the introduction of global or local features using either the global or local fusion module alone can improve performance, but the optimal results can only be achieved when the global and local fusion modules work together. For Yelp, which is a dataset with dense interactions and features on the target behavior can be learned well using the target behavior interactions. Due to its weak behavior correlations, the introduction of global or local features by using the global or local fusion module alone has less effect, but the performance of Yelp can be stabilized at optimal results when using both fusion modules. These results confirm the complementarity between the global and local fusion modules, together with their generalization and robustness across diverse datasets.

3) *Auxiliary Behavior Effectiveness Analysis*: In order to verify the effectiveness of the auxiliary behaviors, we perform ablation experiments on the auxiliary behaviors. Specifically, we remove different auxiliary behaviors at a time and observe how the recommendation performance changes in the presence of reduced auxiliary behaviors. The experiment results are shown in Fig. 5. We can clearly find that in all datasets, the absence of any auxiliary behaviors leads to a decline in recommendation performance, which indicates that our proposed method fully exploits all auxiliary behaviors, and each of them contributes to the representation learning of the target behavior. For instance, in the Taobao dataset, removing either *cart* or *click* behavior significantly reduces both HR@10 and NDCG@10 metrics. Similarly, in the Tmall dataset, the absence of *click* behavior notably decreases the performance, especially for HR@10. This is due to the fact that this behavior has the most intensive number of interactions and plays a significant role in mitigating the interaction sparsity of the target behavior. In the Yelp dataset, the removal of either *negative* or *tips* behavior also leads to a noticeable drop in both metrics. These results suggest that each type of auxiliary behavior plays a crucial role in enhancing the recommendation system’s performance across different datasets.

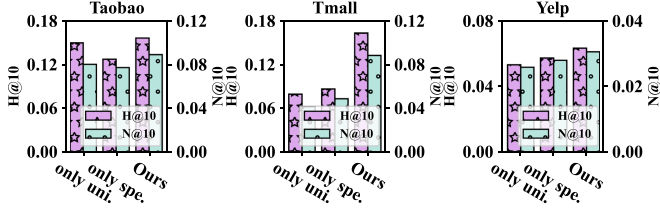


Fig. 6. Impact of fusion or non-fusion between behavior-unified and behavior-specific features.

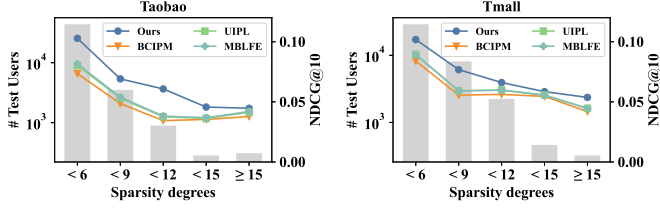


Fig. 7. Performance comparison w.r.t different data sparsity degrees on Taobao and Tmall.

#### D. Impact of Feature Fusion

In this section, to further validate the necessity of fusing behavior-unified features and behavior-specific features, we designed comparative experiments between fusion and non-fusion approaches, where non-fusion corresponds to two variants: (1) **only uni.** : using only behavior-unified features and (2) **only spe.** : using only behavior-specific features. The results of the comparative experiments are shown in Fig. 6. Consistently, the experimental results demonstrate that fusion achieved the optimal performance across all three datasets, while non-fusion, whether using only behavior-unified features or only behavior-specific features, results in a performance degradation. This suggests that relying solely on behavior-unified features or behavior-specific features alone cannot fully exploit the potential of multi-behavior recommendations, and both should be given attention for optimal performance. Furthermore, we observe that the performance of utilizing only behavior-unified features or behavior-specific features exhibits variability across different datasets. On the Taobao dataset, the performance of employing only behavior-unified features is superior to that of using only behavior-specific features. Conversely, this is not the case on the Tmall and Yelp datasets. This indicates that relying solely on behavior-unified features or behavior-specific features is insufficient to ensure adequate generalization across different datasets as fusion does, further emphasizing the necessity of fusion.

#### E. Robustness to Data Sparsity

We present the comparison results with the some competitive baselines (BCIPM, UIPL and MBLFE) on Taobao and Tmall datasets due to space limitations. The experiment results are shown in Fig. 7. In Fig. 7, the x-axis denotes the different data sparsity degrees, the left side of y-axis displays the number of test users in the corresponding group quantified by bar while the right side of y-axis is the averaged metric value quantified by line. According to Fig. 7, we have the following observations: i) Our model can consistently outperforms BCIPM, UIPL and

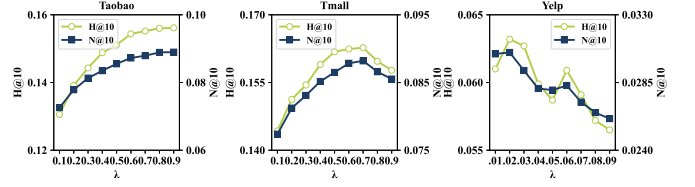


Fig. 8. Impact analysis of optimization trade-off coefficient  $\lambda$ .

TABLE II  
TRAINING TIME COMPARISON (SECONDS PER EPOCH) OF DIFFERENT METHODS ON ALL THREE DATASETS

| Training time (s)<br>Dataset | Model | BCIPM  | MBLFE  | UIPL  | Ours  |
|------------------------------|-------|--------|--------|-------|-------|
|                              |       |        |        |       |       |
| Taobao                       |       | 565.10 | 128.67 | 22.67 | 30.12 |
| Tmall                        |       | 555.98 | 112.67 | 19.33 | 26.29 |
| Yelp                         |       | 459.87 | 49.33  | 13.82 | 20.56 |

MBLFE across different datasets. ii) There is a slight decline in performance as the number of interactions increases, which may be attributed to difference in the amount of available auxiliary data. We take the Tmall dataset for example. On the Tmall dataset, users with more than 15 purchase records have significantly fewer auxiliary behavioral records compared to those with less than 12 purchase records.

#### F. Impact of Hyper-Parameter $\lambda$

Fig. 8 presents the outcomes of our model under different optimization trade-off coefficients  $\lambda$ . From the figure, we can find that as  $\lambda$  increases, the trend of performance varies on different datasets. On the Taobao dataset, the performance gradually improves as  $\lambda$  increases, peaking at around 0.8; on the Tmall and Yelp datasets, the general trend of performance is to first increase and then decrease, peaking at 0.7 and 0.02, respectively. This finding indicates that the contribution of the contrastive loss function to performance varies across different datasets. A significant proportion of the contrastive loss function is observed in sparse datasets such as Taobao and Tmall, as an important way to preference alignment and knowledge transfer in enhancing preference features on the sparse target behavior.

#### G. Efficiency Analysis

We evaluate the computational efficiency of the proposed method by comparing its average training time per epoch with three state-of-the-art MBR methods under identical hardware settings (CPU: Intel(R) Xeon(R) Gold 6133@2.50 GHz, GPU: GeForce RTX 4090, Embedding Size: 64, Batch Size: 1,024). As shown in Table II, the results demonstrate that our method achieves a superior balance between recommendation performance and computational overhead. Compared to high-performing baselines like BCIPM and MBLFE, our model exhibits significantly higher efficiency. Specifically, BCIPM incurs the highest computational overhead due to its complex behavior-contextualized modeling. Similarly, MBLFE is considerably slower than our method, which can be attributed to the heavy computational burden associated with maintaining numerous latent factor expert networks. In contrast, our model achieves a

substantial reduction in training latency, maintaining a speed that is an order of magnitude faster than these complex architectures. This efficiency may stem from our structured fusion design, which avoids the extensive parameterization of multiple experts as in MBLFE or the complex interaction modeling as in BCIPM. When compared with UIPL, which is the most lightweight baseline, our model exhibits a slightly higher but highly competitive training time. While UIPL relies on a relatively static feature initialization strategy to handle behavior-unified global features, our model implements a hierarchical fusion mechanism, which is a justifiable trade-off to capture intricate user preferences across diverse behaviors. Consequently, the results confirm that our model strikes an optimal balance between recommendation performance and computational efficiency.

## V. CONCLUSION

In this study, we reexamine multi-behavior recommendation approaches and introduce a novel multi-behavior recommendation method called UGLP that proposes a behavior feature fusion network that conducts global and local fusion on behavior-unified and behavior-specific features to unify behavior-unified global and behavior-specific local preferences for comprehensive user preference modeling. We further introduce a contrastive learning module to promote preference alignment and facilitate knowledge transfer from auxiliary behaviors to the target behavior. Additionally, we employ a GCN refinement module to prevent the loss of global user preference information. Extensive experiments on three real-world datasets validate our method's efficacy, with ablation experiments confirming the importance of its components. Future work will focus on the improvement of both fusion methods and refinement modules to boost recommender system performance further.

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